Election-Result-Management System

A Data Structure Project Report

Submitted By:

Mayank Raj

Roll No : 25BTCSEAI0204

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in

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by

Under the Guidance of:

Dr. Sumit Sharma

Assistant Professor, SoEC

Department of Computer Science & Engineering

Dev Bhoomi Uttarakhand University

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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CERTIFICATE

This is to certify that the report entitled **Election-Result-Management System** submitted by **Mayank Raj** (Roll No : 25BTCSEAI0204), to Department of Computer Science & Engineeringin partial fulfillment of the B.Tech. degree in **Computer Science & Engineering** is a bonafide record of the seminar work carried out by him under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Dr. Sumit Sharma
(Project Guide)
Assistant Professor
Dept.of Computer Science & Engineering
Dev Bhoomi Uttarakhand University
Dehradun, Uttarakhand

Mr. Bhushan Kumar
(Project Coordinator)
HoD, Assistant Professor
SOEC
Dev Bhoomi Uttarakhand University
Dehradun, Uttarakhand

DECLARATION

I Mayank Raj hereby declare that the project report titled **Election-Result-Management System**, submitted to demonstrate my understanding of data structures and their applications uses, is my original work completed under the supervision of Dr. Sumit Sharma.

This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources.

I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Mayank Raj

Dehradun, Uttarakhand

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Abstract

This document contains essential templates required to write technical reports using LaTeX. This template may be used for the preparation of B.Tech seminar reports of Rajasthan Technical University, Kota, Rajasthan. Also minimum working examples to create equations, include figure, include table, table of contents symbols list and bibliographic citation in a LaTeX document are provided.

Please note that this template is provided without warranty on an AS IS basis.

Acknowledgement

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Mayank Raj

Chapter 1

Introduction

The Election Result Management System is a digital solution designed to address the ineffiencies and security risks inherent in traditional paper-based voting.Conventional systems often suffer from manual counting errors, high resource consumption , and the difficulty of preventing fraudulent activities like duplicate voting.This projects aims to provide a lightweight ,secure,and transparent alternative by leveraging efficient data structures.

1.1 Problem Statement:

In traditional paper-based voting systems, several inefficiencies and security risks exist, including manual counting errors, high resource consumption, and the difficulty of preventing duplicate voting. There is a need for a lightweight, digital alternative that can:

- Automate Tabulation: Eliminate human error in counting votes.
- Enforce Integrity: Ensure that no individual can cast more than one vote.
- Handle Dynamic Participation: Accommodate an unpredictable number of candidates and voters without memory wastage.

1.2 Objective

To design and implement a robust Election Management System using Linked Lists to provide dynamic memory management and efficient data handling for a transparent voting process.

Chapter 2

ER-Diagram

The system is modeled around two primary entities: Candidate and Voter. Although they exist in separate linked lists for security, they interact through the ”Votes For” relationship

Relationship Description:

• Candidate Entity: Stores information about those contesting.

• Voter Entity: Records who has already participated to prevent fraud.

• Interaction: When a vote is cast, the Voter’s ID is logged in the Voter list, and the Candidate’s votes attribute is incremented.

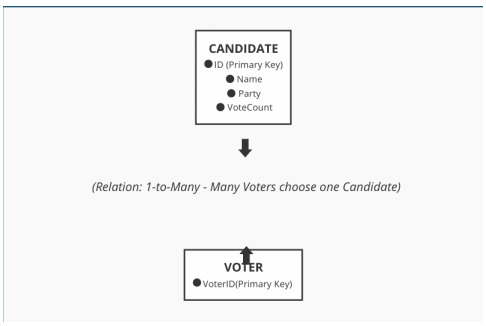

Figure 2.1

Chapter 3

Data Structures Architecture

The project utilizes Singly Linked Lists. Unlike arrays, linked lists allow the system to grow dynamically as new candidates register during the nomination phase.

3.0.1 Node Definition

```
struct Candidate {
    int id;
    char name[50];
    char party[50];
    int votes;
    struct Candidate* next;
};

struct Voter {
    int voterId;
    struct Voter* next;
};
```

Figure 3.1: Node Definition

Chapter 4

Implementation Logic

The system follows a modular functional approach:

1. Candidate Management: Uses isDuplicateCandidate() to traverse the list and check for ID existence before insertion at the tail.

2. Voting Integrity: Before a vote is recorded, hasVoted() performs a linear search on the Voter list. If the ID is not found, a new Voter node is prepended to the head.

3. Winner Detection: Traverses the Candidate list once, tracking the maximum value and checking for ties.

Chapter 5

Results

— Election Management System —

1. Add Candidate
2. Cast Vote
3. Display Results
4. Find Winner
5. Exit

Enter choice: 1

Enter Candidate ID: 1008

Enter Name: Mayank Raj

Enter Party: BJ

Candidate Mayank Raj added successfully!

— Election Management System —

1. Add Candidate
2. Cast Vote
3. Display Results
4. Find Winner
5. Exit

Enter choice: 1

Enter Candidate ID: 108

Enter Name: Tej

Enter Party: RJ

Candidate Tej added successfully!

— Election Management System —

1. Add Candidate

2. Cast Vote

3. Display Results

4. Find Winner

5. Exit

Enter choice: 2

Enter your unique Voter ID: 1008

Enter Candidate ID to vote for: 1008

Vote cast successfully for Mayank Raj!

— Election Management System —

1. Add Candidate

2. Cast Vote

3. Display Results

4. Find Winner

5. Exit

Enter choice: 2

Enter your unique Voter ID: 108

Enter Candidate ID to vote for: 1008

Vote cast successfully for Mayank Raj!

— Election Management System —

1. Add Candidate

2. Cast Vote

3. Display Results

4. Find Winner

5. Exit

Enter choice: 2

Enter your unique Voter ID: 10058

Enter Candidate ID to vote for: 1008

Vote cast successfully for Mayank Raj!

— Election Management System —

1. Add Candidate

2. Cast Vote

3. Display Results

4. Find Winner

5. Exit

Enter choice: 1008

Invalid Choice!

— Election Management System —

1. Add Candidate

2. Cast Vote

3. Display Results

4. Find Winner

5. Exit

Enter choice: 2

Enter your unique Voter ID: 1008

Access Denied: Voter ID 1008 has already cast a vote!

— Election Management System –

1. Add Candidate

2. Cast Vote

3. Display Results

4. Find Winner

5. Exit

Enter choice: 22

Invalid Choice!

— Election Management System —

1. Add Candidate

2. Cast Vote

3. Display Results

4. Find Winner

5. Exit

Enter choice: 2

Enter your unique Voter ID: 102

Enter Candidate ID to vote for: 1008

Vote cast successfully for Mayank Raj!

— Election Management System —

1. Add Candidate

ID	Name	Party	Votes
1008	Mayank Raj	BJ	5
108	Tej	RJ	0

Table 5.1: Election Result

2. Cast Vote

3. Display Results

4. Find Winner

5. Exit

Enter choice: 2

Enter your unique Voter ID: 100

Enter Candidate ID to vote for: 1008

Vote cast successfully for Mayank Raj!

— Election Management System —

1. Add Candidate

2. Cast Vote

3. Display Results

4. Find Winner

5. Exit

Enter choice: 3

— Election Results —

— Election Management System —

1. Add Candidate

2. Cast Vote

3. Display Results

4. Find Winner

5. Exit

Enter choice: 4

WINNER: Mayank Raj from BJ with 5 votes!

Chapter 6

Conclusion

By leveraging linked list data structures, this system provides an efficient way to manage election data without the constraints of static memory allocation. The ER model ensures a clear separation between candidate data and voter authentication records, maintaining system security and performance.

References

Dynamic Memory Allocation:

A technique in C to allocate memory during runtime. We used malloc() to create new nodes for students only when needed, ensuring the hostel system doesn't waste memory.

Self-Referential Structure:

The struct Student is self-referential because it contains a pointer (next) to an instance of itself, which is the foundational block of a Linked List.